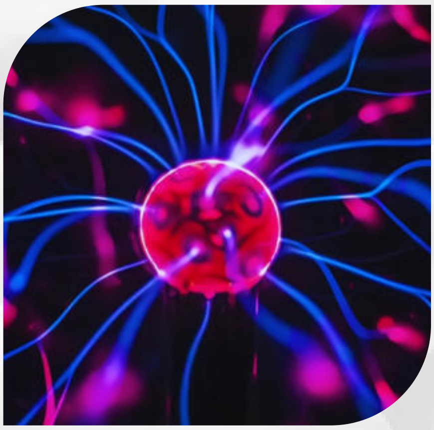
UNIT 14

ELECTROSTATICS



ELECTRIC CHARGE

Electric charge is a fundamental property of matter, defined as a fundamental conserved quantity that can be positive or negative, leading to the attraction or repulsion between charged particles.

PRODUCTION OF ELECTRIC CHARGE

RUBBING A BALLOON ON YOUR HAIR

When you rub a balloon on your hair, some electrons transfer from your hair to the balloon. As a result, the balloon becomes negatively charged, and your hair may gain a positive charge. Due to this charge imbalance, the negatively charged balloon may stick to neutral objects like a wall, or your hair may stand on end due to the repulsion of like charges.

TYPES OF CHARGES

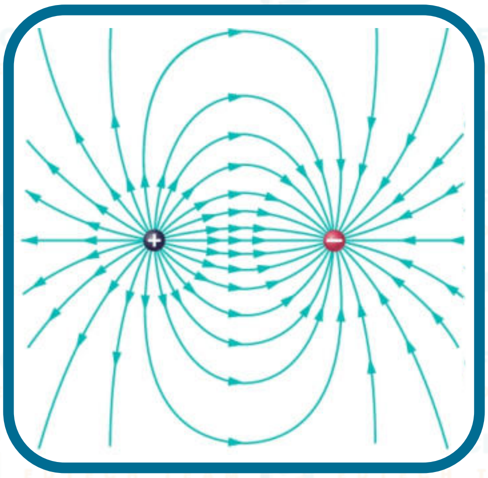
There are two types of electric charge.

POSITIVE CHARGE

Associated with an excess of protons, causing objects to repel other positively charged objects and attract negatively charged ones.

NEGATIVE CHARGE

Resulting from an excess of electrons, leading to repulsion from other negatively charged objects and attraction to positively charged ones.



METHOD OF CHARGE FORMATION

Induction: The process of redistributing electric charge in a conductor without direct contact with a charged object.

Example: Bringing a negatively charged balloon near an uncharged metal can causes electrons in the can to move, leaving one side negatively charged and the other positively charged.

Friction: Charging by rubbing two dissimilar materials together, causing the transfer of electrons between them.

Example: Rubbing a plastic comb with dry hair transfers electrons, making the comb negatively charged, and the hair positively charged.

Conduction: Charging an object by direct contact with a charged object, leading to an equalization of charge between them.

Example: Touching a positively charged metal sphere with a neutral metal rod transfers charge, leaving both objects with equal positive charge.

ELECTROSTATIC INDUCTION

Electrostatic induction is the process of redistributing electric charge in a conductor by bringing a charged object nearby, without any direct contact, leading to a temporary separation of charges within the conductor.

ELECTROSTATIC CHARGING BY INDUCTION

Electrostatic charging by induction is the process of creating a charge imbalance in a neutral object by bringing a charged object close to it without direct contact, resulting in the temporary separation of charges within the neutral object. This occurs due to the influence of the charged object's electric field on the charges in the neutral object.

ELECTROSCOPE

An electroscope is a simple instrument used to detect and indicate the presence of electric charges. It can determine whether an object is charged and, in some cases, the type of charge (positive or negative).

CONSTRUCTION OF ELECTROSCOPE

A basic electroscope consists of the following components:

1. A metal rod or stem:

This is a conducting material, typically made of metal that extends vertically from the base of the electroscope.

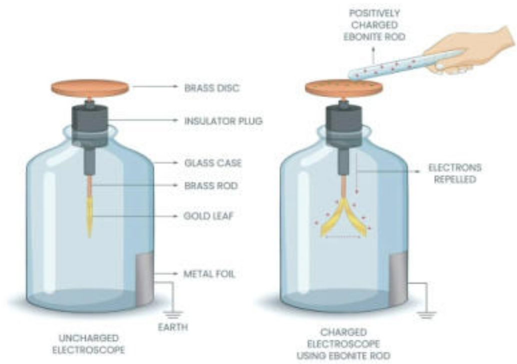
2. A metal disc or knob:

Located at the top end of the metal stem, this disc serves as a terminal for the electroscope.

3. A metal leaf or leaves:

Two thin metal leaves (usually made of gold or aluminum) are attached to the lower end of the metal stem. The leaves are lightweight and have a natural repulsive force when they have the same electric charge.

GOLD LEAF ELECTROSCOPE



1) Working of Electroscope:

When an electrically charged object is brought near the metal knob of the electroscope, the following process occurs:

2) Charging by Induction:

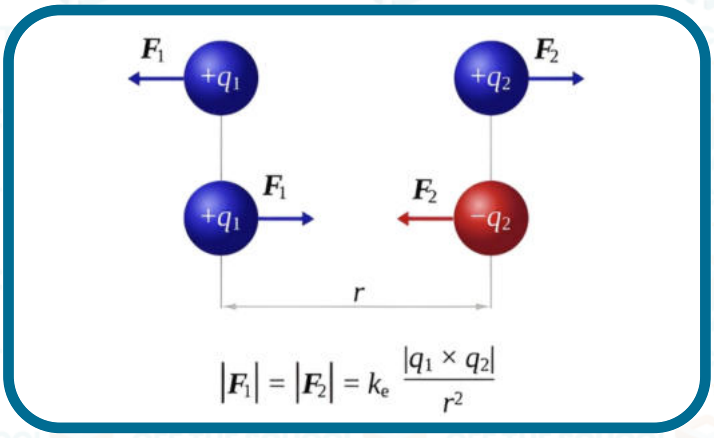
A charged object is brought near the metal knob, causing the redistribution of charges in the electroscope's knob through electrostatic induction.

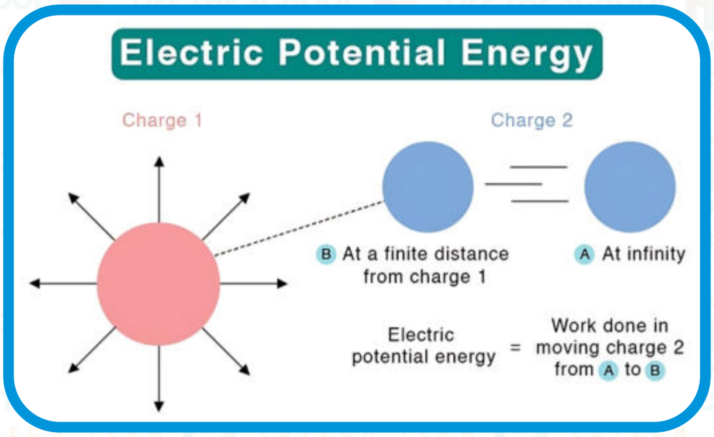
3) Leaf Separation:

The repulsion between like charges (e.g., both being negatively charged) in the metal leaves causes them to diverge or separate.

4) Charge Detection:
The separation of the metal leaves visually indicates the presence of a charge on the electroscope. If the leaves diverge, it suggests opposite charges with the charged object, and if the leaves collapse, it implies similar charges with the charged object.

ELECTROSTATIC CHARGING BY INDUCTION 'S LAW
Coulomb's Law is a fundamental principle in physics that describes the electrostatic force between two charged particles. **It states that the force between two-point charges is directly proportional to the product of their magnitudes and inversely proportional to the square of the distance between them.**





MATHEMATICAL EXPRESSION OF COULOMB'S LAW
Coulomb's Law is mathematically expressed as:

$$F = k\, q_1 q_2 / r^2$$

Where:
- (F) is the electrostatic force between the two charges.
- k is Coulomb's constant, approximately 8.99 * 109 Nm2/C2 (Coulomb's constant depends on the medium in which the charges are present; in a vacuum, it is 8.99 * 109 Nm2/C2)
- q1 and q2 are the magnitudes of the two point charges.
- r is the distance between the centers of the two charges.

ELECTRIC FIELD
Electric field is a region of space around a charged object where another charged particle would experience a force due to the presence of the first charge.

ELECTRIC FIELD INTENSITY

Electric field intensity (also known as electric field strength) at a point in space is the force experienced per unit positive charge placed at that point.

FORMULA FOR ELECTRIC FIELD INTENSITY (E)

$$E = F/q$$

Where:

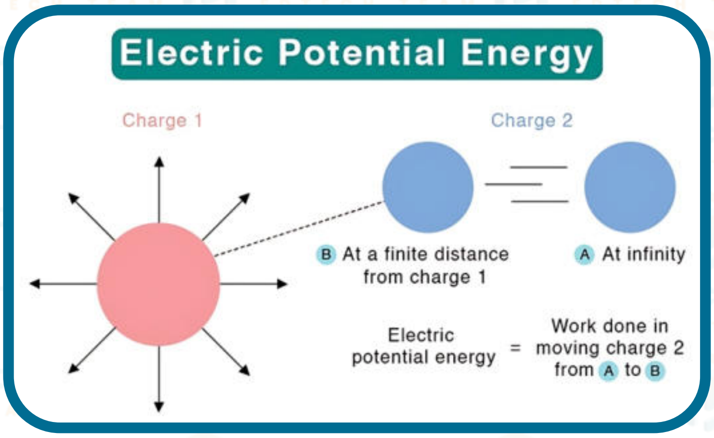
- (E) is the electric field intensity at a point.
- (F) is the force experienced by a positive test charge placed at that point in the electric field.
- (q) is the magnitude of the positive test charge.

ELECTRIC FIELD LINES

Electric field lines are imaginary lines used to visually represent the direction and strength of the electric field around a charged object. They originate from positive charges and terminate on negative charges, showing the path that a positive test charge would follow if placed in the electric field. The density of these lines indicates the strength of the field, with closer lines indicating a stronger field. The lines never intersect, and their direction at any point gives the direction of the electric field at that point.

ELECTROSTATIC POTENTIAL

Electrostatic potential is the amount of work required to move a unit positive charge from infinity to a specific point in an electric field, providing a measure of the potential energy associated with the charge's position.



FORMULA FOR ELECTROSTATIC POTENTIAL (V)

$$V= (k.q)/r$$

Where:

- V is the electrostatic potential at a point.
- k is Coulomb's constant (8.99 * 109 Nm2/C2 in vacuum).
- q is the magnitude of the point charge creating the field.
- r is the distance from the point charge to the point where potential is measured.

VOLT

Volt is the unit of electric potential, representing one joule of energy per coulomb of charge.

Unit:

It S.I unit is Volt (V).

POTENTIAL DIFFERENCE

Potential Difference, also known as voltage, is the measure of the electric potential energy difference between two points in an electric field, indicating the work done per unit charge in moving a charge between those points.

APPLICATION OF ELECTROSTATIC

1. Spray Painting: Electrostatic spray painting is used in industries to achieve even and efficient coating of surfaces. Paint particles are charged and sprayed onto the object to be painted, which is oppositely charged. This attraction leads to better coverage and reduced paint wastage.

2. Electrostatic Air Cleaning: Electrostatic air cleaner's use charged plates or grids to attract and capture airborne particles, such as dust, pollen, and pollutants. As air passes through the device, particles become charged and adhere to oppositely charged surfaces, improving indoor air quality.

3. Inkjet Printing: Inkjet printers use electrostatic forces to precisely control the ejection of ink droplets onto paper, creating high-resolution images and text.

4. Copiers and Laser Printers: Photocopiers and laser printers use electrostatics to transfer toner (charged powder) onto paper, which is then fused onto the paper using heat, creating images and text.

5. Electrostatic Precipitators: These devices are used to remove particles like ash, smoke, and pollutants from industrial exhaust gases by giving them an electric charge and then attracting them to oppositely charged plates.

6. Van de Graff Generator: This electrostatic generator produces high-voltage static electricity for various scientific experiments and demonstrations.

CAPACITOR

A capacitor is an electronic component that stores and releases electrical energy. It consists of two conductive plates separated by an insulating material (dielectric), creating an electric field between them. When a voltage is applied across the plates, one plate accumulates positive charge while the other accumulates negative charge, creating a potential difference.

CAPACITANCE

Capacitance is the measure of a capacitor's ability to store electric charge when a potential difference is applied across its plates.

DERIVATION OF CAPACITANCE FORMULA

The capacitance (C) of a parallel plate capacitor is given by the formula:

$$C = \frac{Q}{V} = \frac{\varepsilon_o A}{d}$$

$$\text{Capacitance} = \frac{Magnitude\ of\ charge\ on\ Conductor}{Magnitude\ of\ potential\ difference}$$

CAPACITANCE DEPENDS ON THESE FACTORS

- Area of the plate. Capacitance increases if area of the plate increases.

Hence C∞ A.

- Distance between the plates. Capacitance increases if the separation distance between the plates decreases.

Hence C∞ 1/d

- Dielectric Constant (C) capacitance increases if insulating medium of high dielectric constant is used.

Hence C ∞ ε_r.

COMBINATIONS OF CAPACITORS

Capacitors can be combined in parallel or in series to achieve different effective capacitance values.

1. **Parallel Combination:** In a parallel combination, the positive plates of all capacitors are connected together, and the negative plates are also connected together.

DERIVATION FOR PARALLEL COMBINATION

Let suppose capacitor consists on such a combination that positive terminal of each capacitor is connected with the positive terminal of the other capacitor and negative terminal of each capacitor is connected with the negative terminal of the other capacitor. Then the combination is said to be Parallel combination.

If three Capacitors C1, C2 and C3 are connected in Parallel and further connects them with a battery of V volts then: C_1 draws charge Q1, C2 draws charge Q2 and C3 draws charge Q3. Then:

$$Q = Q_1 + Q_2 + Q_3$$

By applying Capacitor equation. We get:

$$Q_1 = C_1V,\ Q_2 = C_2V,\ Q_3 = C_3V,\ Q = CV$$

So capacitance becomes:

$$C_{eq}V = C_1V + C_2V + C_3V$$

$$C_{eq}V = (C_1 + C_2 + C_3)\,V$$

$$C_{eq} = C_1 + C_2 + C_3$$

SERIES COMBINATION OF CAPACITORS

2. SeriesCombination: In a series combination, the positive plate of one capacitor is connected to the negative plate of the next capacitor, forming a chain.

DERIVATION FOR SERIES COMBINATION

Let suppose Capacitors are consists on such a combination that positive terminal of one Capacitor connected with the negative terminal of the other Capacitor and the negative terminal of first Capacitor is connected with the positive terminal of the other Capacitor. Then the combination is said to be Series combination.

If three Capacitor C1, Cz and C3 are connected in Series and further connects them with a battery of V volts.

Then: Positive plate of Capacitor C1 draws charge +Q from the battery ad negative plate of C3 draws charge - Q from the battery.

The charge +Q on the positive terminal (Left Plate) of Ci attracts free electrons from the left plate of Cz and these free electrons are accumulated on the right plate of C1. Thus right plate of C_1 becomes negatively charged with a charge-Q. in this way every Capacitor becomes charged. If voltage acquired by each Capacitor is V1, V2, V3 by applying Capacitor equation on C1, C2 and C3. We get:

$$Q = C1V1,\ Q = C2V2,\ Q = C3V3,\ Q = C_eV$$

$$V1 = \frac{Q}{C1},\ V2 = \frac{Q}{C2},\ V3 = \frac{Q}{C3}$$

$$\frac{Q}{Ce} = Q\left(\frac{1}{C1} + \frac{1}{C2} + \frac{1}{C3}\right)$$

$$\frac{1}{Ce} = \frac{1}{C1} + \frac{1}{C2} + \frac{1}{C3}$$

So now according to the equation:

The reciprocal of equivalent capacitance is equal to the sum of reciprocals of individual capacitance.

USES OF CAPACITORS

Capacitors have various applications across multiple fields. Some of their uses include:

- **Energy Storage:** Capacitors store electrical energy and provide quick bursts of power, making them essential components in flashlights, camera flashes, and defibrillators.
- **Tuning and Matching:** Capacitors are employed in tuning radio receivers, antennas, and circuits to resonate at specific frequencies, optimizing performance.
- **Electric Motors:** Capacitors are used in split-phase and capacitor-start motors to control phase relationships and starting performance.
- **Voltage Regulation:** Capacitors stabilize voltage levels, reduce voltage fluctuations, and prevent voltage spikes in electronic systems.
- **Memory Storage:** Capacitors serve as charge storage elements in dynamic random-access memory (DRAM) chips, used in computers and digital devices.
- **Flash Memory:** Capacitors play a role in NAND flash memory cells, storing data in devices like USB drives, SSDs, and memory cards.